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Top Skills

Machine Learning

Data Science

Product Management

Certifications

Strategic Product Leader

FUNDAMENTALS OF
ACCELERATED DATA SCIENCE
WITH RAPIDS

MLx Health & Bio

BUILDING INTELLIGENT
RECOMMENDER SYSTEMS

MLx Representation Learning &
Generative AI

Honors-Awards

CREPUQ Scholarship

Cultural Exchange to Compiègne,
France

2nd Place - Game Expo

Professional Member - MBCS

Winner at the DSS Hackathon 2019 -
"Best Business Solution"

Publications

Improving Inverse Folding for
Peptide Design with Diversity-
Regularized Direct Preference
Optimization

Bruno Trentini

ML Research @ NVIDIA | PhD Student @ University of Oxford | AI +
Digital Bio/Chem
London, England, United Kingdom

Summary

Applied Research Scientist at NVIDIA and PhD Student in the
Computer Science Department of the University of Oxford with Prof.
Michael Bronstein. My research covers Generative Models and
Dynamical Systems for Molecular Design.

Skills

- Python (PyTorch, PyG, DDP, NumPy, Matplotlib, Pandas, SciPy),
Apache Spark,
- Wandb, Databricks, MS Azure
- Machine Learning & Deep Learning (GNN, Transformers, Diffusion
Models, Flow Matching, Graph Neural Networks)

Experience

NVIDIA

3 years 7 months

Applied Research Scientist - Digital Biology
November 2024 - Present (1 year 1 month)

Researching generative AI & dynamics for molecular systems.

I took on this role alongside my PhD at the University of Oxford, which began
in October 2024, to pursue closely aligned research objectives.

Global Research Alliances Lead - Life Sciences

August 2023 - November 2024 (1 year 4 months)

London Area, United Kingdom

- Led cross-institution research collaborations and scientific community
outreach.
- Organised and delivered scientific workshops and tutorials.
- Revised proposals and authored grant letters of support.
- Presented NVIDIA products and open-source tools to research audiences.

- Tracked state-of-the-art methods; operationalised NVIDIA tooling for researchers.
- Research Topics: Peptide Design, protein inverse folding; LLMs for biologics.

Developer Relations Manager - EMEA Healthcare & Life Sciences Startups Lead

May 2022 - August 2023 (1 year 4 months)

London, England, United Kingdom

- Organised scientific workshops and community sessions; presented NVIDIA products and open-source tools to research audiences.
- Led engagements on the Cambridge-1 supercomputer; delivered outcomes including Alchemab's Fabcon (bioRxiv preprint, 2024) (<https://www.biorxiv.org/content/10.1101/2024.05.22.594943v3.full.pdf>)
- Trained 2,000+ developers through developer summits, online meet-ups, and local conferences
- accelerated research pipelines and production workloads using GPU-accelerated libraries and cloud APIs; reduced experimentation friction for startups technical staff.
- Contributed to the BioNeMo Framework with demo assets and documentation
- Delivered technical evangelism and hands-on support to the EMEA HCLS startup ecosystem, enabling adoption of NVIDIA platforms and open-source tooling.
- Managed technical alliances and multi-party scientific projects

Selected Achievements (Public Links)

Accelerating Drug Discovery at Receptor.AI with NVIDIA BioNeMo Cloud APIs — NVIDIA Developer Blog (<https://developer.nvidia.com/blog/accelerating-drug-discovery-at-receptor-ai-with-nvidia-bionemo-cloud-apis/>)

Breakthrough in Functional Annotation with HiFi-NN — NVIDIA Developer Blog (<https://developer.nvidia.com/blog/breakthrough-in-functional-annotation-with-hifi-nn/>)

Accelerating Neurosymbolic AI with RAPIDS and Prometheux Vadalogue Parallel — NVIDIA Developer Blog

Highlights from the Healthcare & Life Sciences Developer Summit — NVIDIA Developer Blog (<https://developer.nvidia.com/blog/latest-discoveries-at-the-healthcare-life-sciences-developer-summit/>)

University of Oxford

Casual Teacher

January 2025 - March 2025 (3 months)

Teaching assistant for the Geometric Deep Learning course

GSK

1 year 8 months

Senior Data Scientist

October 2020 - May 2022 (1 year 8 months)

London Area, United Kingdom

- Developed Recommender Systems based on Bayesian Machine Learning.
- Developed contextual bandit systems with Bayesian optimisation. Technical lead on OneID project, shortlisted for DataIQ “Breakthrough with Data” award.
 - Applied treatment effect estimation to quantify variable impact in forecasting and causal inference pipelines;
 - Designed taxonomy, implemented ETL pipelines using Databricks & Apache Spark, engineered features, trained models, deployed APIs.
 - Delivered production interface using Django/Flask for internal adoption.

Product Manager - Analytics

October 2020 - March 2021 (6 months)

London Area, United Kingdom

- Led data science projects and built supporting infrastructure.
- Led 3 data engineers to deliver production pipelines for analytics and ML use cases.
 - Improved poorly defined data science workflows; introduced standards for model development, versioning, and validation.
 - Built data warehouses to support reproducible analysis and scalable modelling.
 - Supported the rollout of analytics infrastructure
 - Worked closely with product and business teams to scope and prioritise data science initiatives.
 - Oversaw development of Power BI dashboards used by global teams for monitoring and reporting business performance across regions

KIT-AR

Machine Learning Research - UCL MSc Thesis

June 2020 - October 2020 (5 months)

London, England, United Kingdom

Machine Learning Research in Graph Representation Learning, Spatio-temporal Systems, and Human Activity Recognition.

NVIDIA

AI Startups Ecosystem Manager

December 2018 - December 2019 (1 year 1 month)

London Area, United Kingdom

Led the NVIDIA Inception Programme for UK & Ireland covering technical evangelism (NVIDIA hardware & software) and engagements with Venture Capital firms.

Hotelbeds

Commercial Analytics Manager

October 2017 - October 2018 (1 year 1 month)

London Area, United Kingdom

Implemented Machine Learning models for churn prediction, forecasting, and anomaly detection. Led the first initiatives covering training and inference using PyTorch and Tensorflow on GPUs.

Alliants

Consultant - Product & Data

October 2014 - October 2017 (3 years 1 month)

London Area, United Kingdom

Led and contributed to various projects across Data Science, Technical Product Management, Database Management, and Mobile Apps development.

GVT

Product Manager

June 2012 - September 2014 (2 years 4 months)

Curitiba, Paraná, Brazil

GVT was one of Brazil's leading telecommunications companies. Acquired by Telefónica.

Developed a "Gesture-Based" Set-Top Box Concept using Microsoft Kinect sensors and the C# SDK

Led the development of a new business, collaborating with external partners on API development and systems integration

Strategic Analysis: Led strategic analyses on technology trends, macroeconomic factors, and the telecommunications sector, providing insights that informed executive decision-making.

Started as a trainee in the Operations Department (only 17 selected from 7,000+ applicants nationwide). I analysed data from regional team to support decision-making and managed project for continuous improvements in the Quality control / CRM. I moved to the Innovation Department permanently after a job rotation.

EY

Trainee of Advisory

June 2011 - June 2012 (1 year 1 month)

Developed ACL, SQL, and Shell scripts to detect financial misconduct in database and operating systems. Assisted in auditing systems for financial reporting, following COBIT, ITIL, and Sarbanes-Oxley requirements.

HSBC

2 years 7 months

Trainee Software Analyst

April 2010 - December 2010 (9 months)

Developed COBOL and DB2 applications in a global Customer Data Management programme. Engineered solutions for data orchestration across global systems.

Intern

June 2008 - April 2010 (1 year 11 months)

Simplified terminals based in REXX/CICS in IBM z/OS Mainframe Systems. Developed a technical support web page that reduced the number of level 1 requests to the team of specialists. Received extensive training on Mainframe Technologies as well as scripting and programming languages (Java, C, COBOL, CICS, REXX, DB2, Shell).

Education

University of Oxford

Doctorate - PhD, Machine Learning · (October 2024 - March 2028)

UCL

Master of Science - MS, Data Science & Machine Learning · (2019 - 2020)

Federal University of Technology - Parana

PgDip, Computer Systems Networking and Telecommunications · (March 2012 - September 2013)

Pontifícia Universidade Católica do Paraná

Bachelor of Science - BS, Information Systems · (January 2008 - December 2011)